

SRE

→ Engineering [Cost
Availability
Maintenance

→ Digital Convergence

Sharing information with each other

For example Urey Launching phone which
have suitable result sharing info.
to other companies to make
a suitable perfect.

SRE

Functional Requirement (what is a thing & how it's work)
related to the feature of software

Non-functionality requirement (related req.)
related to security validation
need of user

Req. Engineering — analyze
— Documentation

system req.

set of hardware to
run on a system/
platform

req. of software & hardware
to run a system

Domain req. (Extra)

software req.

set of instruction
to run a application

The process of gathering
understanding, refining
prioritizing and validating
everything project/project
must need to success.

Requirement Process

- ① Identify (stakeholder)
- ② Establish goals & objectives or Feasibility study
- ③ Elicit the requirement (Rizwan Then us now)
- ④ Documentation of requirement
- ⑤ Confirmation of requirement
- ⑥ Prioritize the req.

SRE

levels/layers of Requirements

- 1) Business requirement (Why) (Taking demand of stakeholders)
 → User ^{software} requirement (How) (Expectation)
 → Functional requirement (What) (Feedback)

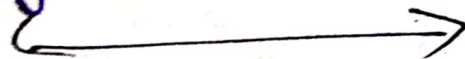
BR (User)	U.R (User)	S.R (Developer)
Outline measurable goals & objectives	Reflect specific needs & expectation of user from the system	Identify FRs, NFRs & use case
Define why making project	Describe who & project (How used & which people are used)	Describe how a product is used
Match project goals with stakeholder goals	Describe how user can interact with it	Describe product (software) as functional a module & non-functional (Babbar Azam) attributes
Maintain BRD with requirements, update arrange	create URs and make them them part of BRD	compose SRs & optionally FRs

SER

Assignment NO. 01
Write a brief note on
software requirements in the
context of system Engineering?

→ Reliability
→ Portability
→ Security

→ constraints
→ Maintenance



2. Req. Engineering in the context of system engineering

System Engineering refers to the specification and functionalities that the software component of a system must fulfil to satisfy the needs of stakeholders.

1. Reliability (perform software function consistently and accurately over time & under various conditions)

2. Constraints (are limitations or restrictions that impact the development process and outcome like time, budget, ^{stakeholder} needs)

Define what can and cannot be achieved within the project scope

3. Portability: refers to the ability of a software application or system to be easily transferred to different environments/platform without requiring significant modifications. A portable system can run effectively and efficiently on various OS.

4. Maintenance (updating/modifying/improving product after its initial development and deployment)
fixing bugs, adding new features

5. Security (protection of the software system from unauthorized access, malicious attack and other security threats. These req. encompass A/A encryption, data integrity).

Requirement Evaluation

- Feasibility study about evolution
- Deal with req. according to their professional
- Changes is made after informed whole teams stakeholders
- If no evolution occurs after affecting teams stakeholders avoid it.

Tips

- All changes must follow the process.
- No implementation or feasibility study if thing not follow process.
- Every changes have record.
- Traceable: Every module is link with others.

Control process ^{feature}

- told which value is add
- How its cost life-cycle
- Healthy conversation is essential

Elobrate: Taking record of feature

scalability: Is it complete in time

informal activity: Non-functional req.

Requirement Traceability

Traceability is a combination of two words. Trace means to find something/someone and ability means capability/skills to do something. So,

Requirement Traceability is the ability to trace a requirement forwards and backwards in the development lifecycle.

Goals

- Provide better quality of product
- find any risk → Easier impact analysis
- keep & verify the record of history and production of an item (documentation)

Types of RT:

① Forward Traceability: (origin)
tracks requirement from the beginning to the end (implementation)

② Backward Traceability:
trace how project element relate to initial requirements. trace requirement from their implementation back to their

origin. It helps in understanding why certain decisions were made during implementation phase by linking them back to the initial requirements.

Bidirectional traceability:

Ability to trace and link requirements both forward and backward. We can track how each requirement is connected to other requirements ensuring that changes made in one area are reflected in all related areas. It helps maintain consistency and ensure that all project elements are aligned.

Requirement traceability matrix

RTM is a way to document the traceability of req. It is a spreadsheet that tracks req. through the development process, collecting data about the req. related coding, the testing has been done and the original reason to include the req. and in the same place.

SRE

Risk

Risk is an uncertain event or condition that if occurs has positive or negative effect/impact on a project.

Risk management

It is the process of identifying, assessing and mitigating risks to minimize their impact on project/organization.

- Proactive risk management
it involves identifying and mitigating potential risks before the occur. of ten through strategies like risk assessment, planning and prevention measures.
→ minimizing likelihood & impact of risk.

- Reactive risk management

It deal with responding to risk after they have happened, typically through strategies like damage control, recovery efforts.

→ focus on minimizing damage and resuming ^{normal} operations after a risk event.

Risk analysis

It is a component of risk management. Risk analysis involves identifying and assessing risk by analyzing their likelihood and potential impact.

Component

1. Risk Identification:

Identifying potential risk that could affect the project. This involves brainstorming sessions, interviews with stakeholders, reviewing historical data and using various techniques such as checklist, flowchart to identify both internal and external risks.

2. Risk Assessment:

Once risks are identified, they need to be assessed to understand their likelihood and potential impact. This involves analyzing factors such as probability of occurrence and timeframe in which they could occur.

Risk assessment often utilizes qualitative or quantitative methods such as probability distribution or statistical model to evaluate risks and prioritize them based on their significance.

3. Risk Evaluation:

It is process of figuring out how serious a risk is and how it could affect our plan. It involves looking at different factors such as how risk is happen and how much it cause damage if occur.
e.g. planning outdoor event, rain came

Risk evaluation helps us to understand risk so we can decide how to deal with it.

4. Risk Mitigation:

Risk mitigation involve steps to reduce the impact of risk. It could involve avoiding the risk, transferring it to another party, reducing its likelihood or having a contingency plan.

Contingency planning

It involves creating backup plan if original plan does not work out. Be ready for unexpected issues and ready to deal with it.

4. Risk Monitoring & Reviews:
Risk analysis is an ongoing process that is important to continuously monitor the identified risks, assessing how ones that may occur and review the effectiveness of the mitigation strategies in place.

Techniques to deal with risk

Brainstorming:

Team members gather to generate ideas and identify potential risk related to software requirements.

Participants freely express their thought on possible risks without judgment, encouraging creativity and diverse perspectives.

Interviews:

Stakeholders including users, developers and project managers are individually interviewed to gather insights (info. collection) on potential risk.

Open-ended questions help elicit (extract) information about concerns, expectations and uncertainties related to requirements.

Checklists:

Pre-defined lists of common risks and issues are used to systematically review requirements.

Team members go through a checklist, marking off potential risks that may apply to the project.

SWOT Analysis

Description: - Examines internal and external factors influencing project success or failure.

How it works: - Identifies strength and weakness within the project team and opportunities and threats from the external environment.

Risk workshops

Involves a structured, facilitated session where team members collectively identify and assess risks.

How it works: Facilitator's guide the team through discussions, ensuring active participation and collaboration in risk identification and assessment.

Use case analysis:

Focuses on analyzing use cases to identify potential risks associated with functional requirements.

How it works: Examine how user interact with the system, helping identify potential pitfalls or misunderstandings in requirements.